

Usage Guidelines

Usage Scenario

A hybrid interface can connect to either a host or a switching device. A hybrid interface allows tagged or untagged VLAN frames to pass.

- To allow a hybrid interface to send tagged VLAN frames, run the **port hybrid tagged vlan** command to add a hybrid interface to VLANs. When the hybrid interface sends frames from these VLANs, it does not remove the VLAN tags. This facilitates communication between switching devices.
- When a hybrid interface connects to a host, run the **port hybrid untagged vlan** command to configure the interface to send untagged frames because the host cannot process tagged frames.

Prerequisites

If the interface is not a hybrid interface, run the **port link-type** command to change the interface type to hybrid.

Configuration Impact

If the **port hybrid untagged vlan** command is run more than once on the same interface, all the configured VLANs take effect on the interface.

Precautions

The command cannot be configured on a physical interface that has been added to an Eth-Trunk interface.

Example

Add the hybrid interface 100GE1/0/1 to VLANs 2, 4, and 50 to 100. The hybrid interface removes these VLAN tags when sending the VLAN frames.

```
<HUAWEI> system-view
[~HUAWEI] interface 100GE 1/0/1
[~HUAWEI-100GE1/0/1] portswitch
[~HUAWEI-100GE1/0/1] port link-type hybrid
[*HUAWEI-100GE1/0/1] port hybrid untagged vlan 2 4 50 to 100
```

2.4.3.32 port link-type

Function

The **port link-type** command configures a link type on an Ethernet interface.

The **undo port link-type** command restores the default link type.

The default link type of an interface is access.

Format

For CE8855, CE8851-32CQ4BQ, CE9865, CE9855, CE8865, CE8875, CE8865-SAN, CE8851-32CQ8DQ-P, CE8850-HAM, CE8850-SAN, CE8851K, CE6866, CE6860-HAM, CE6860-SAN, CE6866K, CE6820H, CE6820H-K, CE6820S, CE6863H, CE6863H-K, CE6881H, CE6881H-K, CE6885-SAN, CE6855-48XS8CQ, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-LL (low latency mode):

port link-type access

port link-type hybrid

port link-type trunk

undo port link-type

For CE8855, CE8851-32CQ4BQ, CE9865, CE9855, CE8865, CE8875, CE8865-SAN, CE8851-32CQ8DQ-P, CE8850-HAM, CE8850-SAN, CE8851K, CE6866, CE6860-HAM, CE6860-SAN, CE6866K, CE6820H, CE6820H-K, CE6820S, CE6863H, CE6863H-K, CE6881H, CE6881H-K, CE6885-SAN, CE6855-48XS8CQ, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L:

port link-type dot1q-tunnel

Parameters

Parameter	Description	Value
access	Configures the link type of an interface as access.	-
hybrid	Configures the link type of an interface as hybrid.	-
trunk	Configures the link type of an interface as trunk.	-
dot1q-tunnel	Configures the link type of an interface as dot1q-tunnel. NOTE This parameter is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6820H, CE6820H-K, CE6820S, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.	-

Views

Layer 2 100GE interface view, Layer 2 10GE interface view, Layer 2 200GE interface view, 25GE-L2 view, 400GE-L2 view, Layer 2 50GE interface view, Layer 2 Eth-Trunk interface view, Interface group view

Default Level

2: Configuration level

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Usage Scenario

Different types of interfaces are applicable to different scenarios:

- An access interface connects to a user host and is usually used on an access link. The frames transmitted on the access link are untagged Ethernet frames.

If an access interface is configured with a default VLAN, the interface adds a tag to the packet and sets the VID in the tag to the default VLAN ID.

- A trunk interface is used to connect to another device and is usually used on a trunk link. A trunk interface allows frames from multiple VLANs to pass through.
- A hybrid interface can connect to a user host or another device. A hybrid interface can connect to an access link or a trunk link. A hybrid interface allows frames from multiple VLANs to pass through.
- A Dot1q-tunnel interface is mainly used on carrier networks. It adds another 802.1Q tag to 802.1Q tagged packets to expand VLAN space. In this manner, packets from private VLANs can be transparently transmitted to the public network.

Precautions

The port link-type command cannot be configured on physical interfaces that are added to an Eth-Trunk.

A Layer 3 interface must be switched to a Layer 2 interface using the **portswitch** command.

When you modify the link type of an interface that has VLAN configuration, the VLAN configuration will be deleted.

Example

```
# Configure the link type of 100GE1/0/1 as trunk.
<HUAWEI> system-view
[~HUAWEI] interface 100GE1/0/1
[~HUAWEI-100GE1/0/1] portswitch
[~HUAWEI-100GE1/0/1] port link-type trunk
```

2.4.3.33 port mux-vlan enable

Function

The **port mux-vlan enable** command enables the MUX VLAN function on an interface.

The **undo port mux-vlan enable** command disables the MUX VLAN function on an interface.

By default, the MUX VLAN function is disabled on an interface.

NOTE

This command is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6820H, CE6820H-K, CE6820S, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.

Format

port mux-vlan enable { vlan { *vlan-id1* [to *vlan-id2*] } &<1-10> }

undo port mux-vlan enable [vlan { *vlan-id1* [to *vlan-id2*] } &<1-10>]

Parameters

Parameter	Description	Value
<i>vlan-id1</i>	Specifies the start VLAN ID.	The value is an integer ranging from 1 to 4094.
to <i>vlan-id2</i>	Specifies the end VLAN ID.	The value is an integer ranging from 1 to 4094.

Views

Layer 2 100GE interface view, Layer 2 10GE interface view, Layer 2 200GE interface view, 25GE-L2 view, 400GE-L2 view, Layer 2 50GE interface view, Layer 2 Eth-Trunk interface view, Interface group view

Default Level

2: Configuration level

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The MUX VLAN function provides a mechanism for isolating Layer 2 traffic between VLAN interfaces. The MUX VLAN function provides principal VLANs and subordinate VLANs. Subordinate VLANs consist of separate and group VLANs. Interfaces in a principal VLAN can communicate with any other interfaces in a MUX VLAN. Interfaces in different subordinate VLANs cannot communicate with each other. Interfaces in a group VLAN can communicate with each other. Interfaces in a separate VLAN cannot communicate with each other.

The preceding functions (isolation or communication) can be implemented only after the MUX VLAN function is enabled.

Configuration Impact

The interface enabled with the MUX VLAN function cannot be added to other VLANs.

Precautions

Access interfaces can be added to only one MUX VLAN group. Trunk and hybrid interfaces can be added to multiple MUX VLAN groups.

Example

```
# Enable the MUX VLAN function on 100GE 1/0/1.  
<HUAWEI> system-view  
[~HUAWEI] vlan 10  
[*HUAWEI-vlan10] mux-vlan  
[*HUAWEI-vlan10] quit
```